Construction Chemicals

SUBMITTAL DOCUMENT

Mixseal PU

One-Component Polyurethane
Joint Sealant & Elastic Adhesive



Elongation	Adhesion	Resistance	Under Water
700 %	Usually no primer	Chemicals & jet fuel	Submerged joints

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Mixseal PU — One-Component Polyurethane Joint Sealant & Elastic Adhesive

Project Name		**Document No.**	OMX-MS-MIXSEALPU
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Mixseal PU	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Mixseal PU on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers joint preparation, installation of backer rod, application and tooling of Mixseal PU one-component polyurethane sealant in expansion and connection joints of buildings, floors, façades, tanks and paved areas.

- Inspection and cleaning of joints.
- Installation of Back-fill backer rod.
- Masking and priming where required.
- Application and tooling of the sealant.
- Curing, protection and inspection of sealed joints.

3. Definitions

Term	Meaning
Skinning time	Time after which a skin forms and the sealant can no longer be tooled.
Tooling	Pressing and smoothing the sealant surface to ensure contact with joint faces.
Backer rod	Foam rod that controls sealant depth and acts as a bond breaker.
Width : depth ratio	Ratio of sealant width to its depth (1 : 1 to 2 : 1).

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.

Role	Responsibility
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Mixseal PU	One-component PU sealant, 600 ml sausages, colour as specified
Back-fill	Closed-cell PE backer rod
Primer	Where required by adhesion tests
Masking tape	For neat edges

5.1 Key product data

Property	Typical Value
Type	One-component, moisture-curing polyurethane
Colours	Black, white, grey
Density	1.15 g/cm³
Hardness	Shore A 40
Modulus at 100 %	approx. 0.3 MPa
Elongation at break	700 %
Adhesion	> 2.0 N/mm² (20 kg/cm²)
Tack-free time	2.5 – 3.5 hours

6. Tools & Equipment

- Sausage gun (manual or pneumatic)
- Knife
- Wire brush
- Blower
- Tooling spatula
- Soapy water
- Depth gauge
- Thermo-hygrometer and surface thermometer (dew point check)

- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Safety glasses
- Safety shoes

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +25 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Measure joint widths and confirm the width : depth ratio.
2. Clean joints of dust, laitance, old sealant, oil and loose material by wire brushing and blowing.
3. Ensure joint faces are dry and sound.
4. Insert Back-fill backer rod to the correct depth.
5. Apply masking tape along both edges.
6. Prime joint faces where adhesion tests show it is needed and allow to dry.
7. Obtain QA/QC approval of prepared joints before sealing.

10. Preparation of Material

1. Check the batch number, colour and expiry date and record them.
2. No mixing is required.

3. Cut the end of the sausage and fit the nozzle cut to suit the joint width.
4. Load the sausage into the gun.

11. Application

1. Extrude the sealant into the joint in one continuous pass, pushing the bead ahead of the nozzle.
2. Fill the joint fully in contact with both faces, avoiding air entrapment.
3. Tool the surface to a slightly concave profile within the skinning time using soapy water.
4. Remove the masking tape immediately after tooling.
5. Record the length of joints sealed and sausages used.

12. Curing & Protection

- Tack-free time 2.5 – 3.5 hours; curing rate 2 – 3 mm per day.
- Protect from water, dust and traffic until cured.
- Barricade sealed floor joints until cured.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Joint faces	Before sealing	Clean, dry, sound	QA/QC
Dimensions	Before sealing	Width 4 – 30 mm, W : D 1 : 1 to 2 : 1	Site Engineer
Backer rod	Before sealing	Correct diameter and uniform depth	QA/QC
Sealant	After tooling	Full contact, no voids, concave finish	QA/QC
Cure	After 7 days	Fully cured, adhered to both faces	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, colour, batch, expiry	Each delivery	Material inspection report
Joint preparation	Clean, dry, width, depth	Each joint	Inspection request (hold point)
Backer rod	Diameter and depth	Each joint	Inspection request
Sealing	Profile, adhesion, finish	Each joint	Daily report
Adhesion test	Pull test after cure	Per area as agreed	Test record

14. Defects & Remedies

Defect	Likely Cause	Remedy
Loss of adhesion	Dirty or wet joint faces; no primer	Cut out, clean, prime and reseal

Defect	Likely Cause	Remedy
Bubbles	Application on wet surfaces; punctured backer rod	Cut out and reseal on dry joint
Cracking in the middle	Sealant too thick; no backer rod	Reseal at correct width : depth ratio
Uneven finish	Late tooling	Tool within skinning time

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact (isocyanates)	Gloves, safety glasses
Work at height	Approved access, harness where required
Cuts from knives	Cut-resistant gloves
Slips and trips	Housekeeping; barricades

- Wear gloves and safety glasses.
- Uncured material contains isocyanates and may cause sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove uncured sealant with a cloth, then wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with a suitable solvent before the sealant cures; cured material can only be removed mechanically.

17. Limitations

- Do not apply below +5 °C or on wet joints.
- Joint width 4 – 30 mm.
- Not for structural glazing.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Joint dimensions verified	☐ Yes ☐ No	
Joints cleaned and dry	☐ Yes ☐ No	
Backer rod installed	☐ Yes ☐ No	
Edges masked	☐ Yes ☐ No	
Primer applied where needed	☐ Yes ☐ No	
Sealant applied without voids	☐ Yes ☐ No	
Tooled within skinning time	☐ Yes ☐ No	
Masking removed	☐ Yes ☐ No	
Protected during cure	☐ Yes ☐ No	
Adhesion checked	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Mixseal PU — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Jointing and Tiling

Mixseal PU

One-Component Polyurethane
Joint Sealant & Elastic Adhesive



Product Description

Mixseal PU is a one-component, moisture-curing polyurethane sealant and elastic adhesive. It seals vertical and horizontal joints, including joints under water, resists chemicals and jet fuel, and in most cases adheres without a primer. It also bonds glass, ceramics, aluminium and concrete elastically.

Uses

- Expansion and connection joints in buildings, floors and façades.
- Joints in water tanks, channels and wet areas.
- Joints in airport aprons and fuel areas.
- Joints around windows, doors and precast panels.
- Elastic bonding of glass, ceramics, aluminium and concrete.

Advantages

- High elongation: 700 %.
- Strong adhesion, usually without primer.
- Resistant to chemicals and jet fuel.
- Suitable for joints under water.
- Non-sag: vertical and horizontal joints.

Application

1. **Joint Preparation**
   Clean, dry, sound joint faces; insert Back-fill backer rod to the correct depth; mask edges.
2. **Gun Application**
   Cut the sausage, load the gun and extrude the sealant into the joint in full contact with both faces.
3. **Tooling**
   Tool the surface within the skinning time using soapy water; remove the masking tape.

Consumption

6 m per 600 ml sausage
for a 10 × 10 mm joint
Consumption (m) = 600 / (width × depth in mm).

Packaging

600 ml sausages.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +25 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Type	One-component, moisture-curing polyurethane	—
Colours	Black, white, grey	—
Density	1.15 g/cm³	—
Hardness	Shore A 40	—
Modulus at 100 %	approx. 0.3 MPa	—
Elongation at break	700 %	—
Adhesion	> 2.0 N/mm² (20 kg/cm²)	—
Tack-free time	2.5 – 3.5 hours	—
Curing rate	2 – 3 mm / day	—
Joint width	4 – 30 mm	—
Width : depth ratio	1 : 1 to 2 : 1	—
Min. application temperature	+5 °C	—

Consumption

Joint size (W × D)	Length per 600 ml sausage
6 × 6 mm	approx. 16 m
10 × 10 mm	approx. 6 m
15 × 10 mm	approx. 4 m
20 × 10 mm	approx. 3 m
30 × 15 mm	approx. 1.3 m

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Do not apply below +5 °C.
- Joint faces clean, dry, sound and free of oil, dust and laitance.
- Joint width 4 – 30 mm; width : depth ratio 1 : 1 to 2 : 1.
- Use Back-fill backer rod to control depth and prevent three-sided adhesion.
- Concrete at least 28 days old.

Application Notes

- Prime porous or critical substrates where adhesion tests show it is needed.
- Mask joint edges for a neat finish.
- Tool with soapy water within the skinning time.
- Allow full cure before exposing to water or traffic; cure rate 2 – 3 mm per day.
- Do not use as a structural glazing sealant.

Health & Safety

- Wear gloves and safety glasses.
- Uncured material contains isocyanates and may cause sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove uncured sealant with a cloth, then wash with soap and water.
- Inhalation: move to fresh air.
- Clean tools with a suitable solvent before the sealant cures; cured material can only be removed mechanically.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

Mixseal PU — Submittal Document

To be attached: approvals and certificates, where available.